

B-helixqa

B — HelixQA submodule update + test-bank authoring survey

Revision: 1 **Last modified:** 2026-06-10T00:00:00Z **Scope:** BobaLink browser-extension feature — HelixQA QA coverage planning **Author:** research subagent (relayed to conductor)

NOTE: The constitution mandates no force-push (§11.4.113) and fetch-first (§11.4.71). The HelixQA pointer bump in the Boba parent repo is **staged in the working tree but NOT committed** — the conductor commits it as part of the planned work.

1. HelixQA submodule update (fetch + pull)

- **Path:** submodules/helixqa
- **Remotes (all fetched OK):** origin = git@github.com:HelixDevelopment/HelixQA.git, GitHub (HelixDevelopment), plus GitLab / VasicDigitalGitHub / VasicDigitalGitLab mirrors (vasic-digital).
- **Default branch:** main (git remote show origin → HEAD branch: main). (A master branch also exists but is stale/unrelated — main is canonical.)

	commit	describe
Before	bcac23639082fa249d716f47a27d86ecc0728428	v4.0.0-385-gbcac236
After	4d2dcb27f87b49e64b54b860689b982079f39c55	v4.0.0-393-g4d2dcb2

- **How far behind:** 8 commits behind origin/main; 0 ahead → clean fast-forward (no divergence).

- **Pull:** `git pull --ff-only origin main` succeeded as a fast-forward.
- **Post-pull verification:** working tree **clean** (`git status --porcelain` empty); **no conflict markers** in `banks/`, `cmd/`, `pkg/`.
- **Parent repo:** Boba now shows `submodules/helixqa` pointer moved `bcac236..4d2dcb2` (staged, **not committed** per instruction).

The 8 new commits (directly relevant — they add the HTTP-bridge run mode):

```
4d2dcb2 test(banks): HXC-048 — pin HXC-SYS-011 wrong-method status
to 404
f18a5d3 test(banks): add helixcode-system http bank — health/
server-info/...
d6c084d feat(cli): add `helixqa http` bridge to drive http: bank
cases against
    a live server (no browser/LLM)           ← KEY for
BobaLink
c7c8236 docs(governance): backfill §11.4.103-121 (CONST-047)
7a19385 docs(governance): cascade §11.4.140/141 (CONST-047)
4934030 test(banks): 6 more HelixCode self-driving banks (49
cases) (§11.4.98)
618dd50 docs(governance): cascade constitution §11.4.122-139
(CONST-047/049)
led8abf test(banks): add foundational HelixCode self-driving test
banks (§11.4.98)
```

2. Test-bank authoring format + exact paths

Where banks live

- All banks: `submodules/helixqa/banks/<name>.yaml` (85 YAML banks on disk).
- **Boba already has banks here** (precedent to copy from):
 - `banks/boba-frontend.yaml` — Angular dashboard (Playwright, web platform)
 - `banks/boba-services.yaml` — backend APIs (`curl/jq`, api platform)
 - `banks/boba-boba-ctl.yaml`, `banks/boba-download-proxy.yaml`, `banks/boba-docs-chain.yaml`
- Filename convention: `kebab-case <purpose>-<scope>.yaml`. An optional JSON peer (`<name>.json`) may share the base name for tooling auto-pairing — not required.

YAML schema (verified against README “Test-bank conventions” + live banks)

Required top-level keys: version (“1.0”), name, test_cases[]. Common optional: description, metadata{author, app, version}.

Per test_case: id (PREFIX-NNN), name, category (functional/performance/security/ux/chaos/...), priority (critical/high/medium/low), platforms (subset of [android, android_tv, web, desktop, ios, aurora_os, harmony_os]; banks also use api), steps[] (each {name, action, expected}), tags[], optional documentation_refs[] / fix_reference{bug_id, description, fix_file}.

Two action styles (pick per how the feature is driven)

(a) http: action — self-driving, NO browser/LLM (NEW, the d6c084d bridge). Best for API endpoints / extension backend. Extra per-step keys expect_status + expect_body_contains make it machine-checkable. Grammar: action: "http: GET /path" or "http: POST /path"; an action: "auth:admin" step logs in first. Reference bank: banks/helixcode-system.yaml, banks/helixcode-auth.yaml.

steps:

- name: "GET /health"
action: "http: GET /health"
expected: "200 OK – body reports status healthy"
expect_status: 200
expect_body_contains: "healthy"

(b) Playwright / curl prose action — browser-driven (the boba-frontend.yaml style). The action/expected strings are read by the LLM-driven autonomous runner; they describe Playwright steps (page.goto(...), page.click(...), page.waitForResponse(...)) or shell (curl ... | jq). This is the path that can actually drive a Chromium UI — relevant because a **browser extension only loads inside a real browser context**.

§11.4.98/§11.4.6: banks describe **structure**, assert machine-checkable outcomes; no hardcoded user-facing prose as the assertion; no guessed JSON paths — assert status + high-confidence substring when shape is unconfirmed.

3. How autonomous QA sessions / banks are run

CLI: submodules/helixqa/cmd/helixqa → bin/helixqa (subcommands in main.go: run, list, report, autonomous, http, replay, signoff, version).

Three run modes:

1. **helixqa http** — **self-driving HTTP, no browser, no LLM** (CI-friendly, the BobaLink-backend path). Exit 0 only if every http: step passed.

```
helixqa http --bank banks/boba-bobalink.yaml \
  --base-url http://localhost:7187 \
  [--admin-user X --admin-pass Y] [--json] [--verbose]
```

--base-url is **required** (never hardcoded, CONST-051(B)). Cases with no http: steps report SKIPPED.

2. **helixqa run** — **structured bank run with platform filter** (uses Playwright for web).

```
helixqa run --banks banks/ --platform web
  # or all / api / android
helixqa list --banks banks/ --platform web      # dry list
```

3. **helixqa autonomous** — **full LLM + vision 4-phase session** (Setup → Doc-Driven Verification → Curiosity Exploration → Report). Drives the app via NavigationEngine (Playwright/ADB/X11), captures screenshots+video, detects issues, emits MD/HTML/JSON report. Needs .env with ≥1 LLM API key.

```
helixqa autonomous --project /path/to/Boba --platforms web \
  --env .env --timeout 30m --output qa-results/ --report
  markdown,html,json
```

4. What to create to add a BobaLink bank

BobaLink is a **browser extension**. Two complementary banks (mirrors the existing boba-services / boba-frontend split):

1. `submodules/helixqa/banks/boba-bobalink.yaml` — the extension's **backend / messaging endpoints** (whatever Boba endpoint the extension calls — e.g. an add-by-URL / search proxy on :7187). Use the **http:** action style (self-driving, CI-gated) with `expect_status + expect_body_contains`. This is the highest-value, no-blocker path.
2. `submodules/helixqa/banks/boba-bobalink-ui.yaml` (optional, browser) — the **in-browser extension UI** (popup/content-script behavior). Use the Playwright prose-action style, `platforms: [web]`, driven by `helixqa run --platform web` or `helixqa autonomous`.
NOTE: a true browser extension must be **loaded into Chromium** (`--load-extension= / chrome.runtime`), which the stock Playwright web path does not set up out-of-the-box — flag this for design (may need a launch-arg hook or an autonomous session configured with the unpacked extension dir).

Skeleton (`banks/boba-bobalink.yaml`):

```
version: "1.0"
name: "Boba BobaLink Browser-Extension Backend Validation Suite"
description: "Self-driving ($11.4.98) validation of the BobaLink
  extension's
  Boba backend surface: <endpoints the extension calls>."
metadata:
  author: "vasic-digital"
  app: "Boba"
  version: "1.0.0"
test_cases:
  - id: BOBA-LINK-001
    name: "BobaLink add-by-URL endpoint accepts a tracker URL"
    category: functional
    priority: critical
    platforms: [api]
    steps:
      - name: "POST <bobalink endpoint>"
        action: "http: POST /api/v1/<bobalink-path>"
        expected: "200/202 – body confirms the link was queued"
        expect_status: 200
        expect_body_contains: "<confirmed-substring>"
    tags: [boba, bobalink, extension, functional]
    estimated_duration: "10s"
    expected_result: "<user-observable outcome>"
    fix_reference:
      bug_id: "BOBA-LINK-ADD-001"
```

```
description: "<...>"
fix_file: "<source path of the handler>"
```

Fill <endpoint>/<substring> from the actual BobaLink design + a real captured response (§11.4.6 — do not guess the JSON shape).

Optionally register the bank in the project's HelixQA Challenge bank / coverage ledger if Boba maintains one (per §11.4.58 / §11.4.27 — none located yet under Boba; confirm with conductor).

5. Blockers / setup notes

- **Go 1.24+** required (helixqa go.work declares go 1.26). make build → bin/helixqa, or go install digital.vasic.helixqa/cmd/helixqa@latest.
- **BLOCKER — missing sibling modules.** helixqa/go.mod has 8 local replace ... => ../<sibling> directives. They resolve to submodules/<sibling>/ in the Boba layout. Present: challenges, containers. **Missing:** doc_processor, llm_orchestrator, llm_provider, llms_verifier (../llms_verifier/llm-verifier), security, vision_engine. Consequence: a full go build ./cmd/helixqa/ **fails** (open ../doc_processor/go.mod: no such file or directory).
 - The http and run modes still need a buildable binary, so this must be resolved before any HelixQA run. Options for the conductor: add the 6 missing repos as siblings (SSH URLs in helixqa/helix-deps.yaml: DocProcessor/LLMOrchestrator/LLMProvider/VisionEngine on HelixDevelopment; LLMsVerifier/security on vasic-digital) via install_upstreams / incorporate-submodule (§11.4.31/§11.4.36), OR build helixqa in its own standalone checkout where the siblings exist and only consume the **bank YAML** from Boba's submodules/helixqa/banks/.
- **autonomous mode** additionally needs .env with ≥1 LLM API key (OpenAI/Anthropic/Google/...), plus VisionEngine (GoCV/OpenCV) and the CLI agent binaries — heavyweight; the **http mode is the lightweight, dependency-light path** and the right first target for BobaLink backend coverage.
- **Browser-extension specificity:** stock helixqa run --platform web drives a plain Chromium page via Playwright; it does **not** auto-load an unpacked extension. Driving the actual extension UI/content-

script needs the extension loaded into the browser (launch args).
Treat as a design item for the UI bank.

Evidence (commands run this session)

- `git -C submodules/helixqa fetch --all --prune` → all 5 remotes fetched OK
- `git -C submodules/helixqa rev-list --count HEAD..origin/main` → 8; `origin/main..HEAD` → 0
- `git -C submodules/helixqa pull --ff-only origin main` → fast-forward to 4d2dcb2; `git status --porcelain` → empty; `conflict-marker grep` → none
- `git -C submodules/helixqa describe --tags` → v4.0.0-393-g4d2dcb2
- `go build ./cmd/helixqa/` → FAIL: missing ../doc_processor/go.mod (+ 5 more)